PSEUDO-COHERENCE OVER A COHERENT RING VIA COHERENCE OF COHOMOLOGY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Over a coherent ring a module is coherent if and only if it is finitely presented. Algebra, Lemma \ref{algebra-lemma-coherent-ring}, cited at proof lines 16564-16565. Identifies two of the three conditions.
- The kernel of a surjection onto a coherent module from a coherent module is coherent. Algebra, Lemma \ref{algebra-lemma-coherent}, cited at proof lines 16566-16568. Lets the resolution by finite free modules be built inductively.
- The cohomology-by-cohomology criterion for m-pseudo-coherence, and the passage from all degrees to pseudo-coherence. Lemma \ref{lemma-cohomology-pseudo-coherent}, Lemma \ref{lemma-n-pseudo-module}, and Lemma \ref{lemma-pseudo-coherent}, cited at proof lines 16574-16579. Give the equivalence of the first two conditions and the final assertion.

Source reference. Original lemma 0EWZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18639–18639. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 05CV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6933–6933. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 05CX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6935–6935. Navigation pointer to a statement the

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Source reference. Original lemma 05CW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6934–6934. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 066B: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7930–7930. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7877–7877. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 064U: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7878–7878. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a coherent ring (Algebra, Definition 05CV). Let $K \in D^-(R)$. The following are equivalent*

1. *K is m-pseudo-coherent,*
2. *$H^m(K)$ is a finite R-module and $H^i(K)$ is coherent for $i > m$, and*
3. *$H^m(K)$ is a finite R-module and $H^i(K)$ is finitely presented for $i > m$.*

Thus K is pseudo-coherent if and only if $H^i(K)$ is a coherent module for all i.

Proof. Recall that an R-module M is coherent if and only if it is of finite presentation (Algebra, Lemma 05CX). This explains the equivalence of (2) and (3). If so and if we choose an exact sequence $0 \to N \to R^{\oplus m} \to M \to 0$, then N is coherent by Algebra, Lemma 05CW. Thus in this case, repeating this procedure with N we find a resolution

$$\ldots \to R^{\oplus n} \to R^{\oplus m} \to M \to 0$$

by finite free R-modules. In other words, M is pseudo-coherent. The equivalence of (1) and (2) follows from this and Lemmas 066B and 064T. The final assertion follows from the equivalence of (1) and (2) combined with Lemma 064U. □